

SeeWeeS Specialty Dispatch Playbook

Multi-Corridor Dispatch Planning (48-hour Horizon)
Version 0.2 — Internal Operations Reference

This document is designed to be machine-readable by automated planning agents.

1. Purpose & Scope

- Defines dispatch planning rules, weather risk logic, and data quality standards.
- Applies to time-critical medicine shipments originating from New Jersey distribution centers.
- Destination: multiple hospital regions across defined delivery corridors.
- Planning horizon: next **48 hours** (Day0 + Day1).

2. System Overview (Narrative Context)

- Medicines are shipped daily from NJ distribution centers across multiple delivery corridors.
- Delivery is time-critical (patient care, cold-chain constraints).
- Dispatch decisions must account for:
 - Weather-driven travel risk
 - Item master data accuracy
 - Truck capacity and packing constraints

3. Corridor Catalog (Authoritative)

The system supports multiple delivery corridors. Corridors are the unit of comparison for risk, KPIs, and resource allocation.

3.1 Corridors

corridor_id	corridor_name	origin_dc	destination_region	default_sla_tier	notes
C1_I95_NJ_BOS	NJ → Boston (I-95)	Newark_NJ_DC	Boston_MA	Tier 1	Existing corridor
C2_NJ_PHL	NJ → Philadelphia	Newark_NJ_DC	Philadelphia_PA	Tier 2	Added corridor for multi-region planning

3.2 Waypoints (per corridor)

Weather risk is evaluated independently at each waypoint. Corridor risk is computed from waypoint risk values across the planning horizon.

C1_I95_NJ_BOS (existing)

waypoint_id	city	lat	lon
C1_W1	Newark NJ	40.7357	-74.1724
C1_W2	Bronx NY	40.8448	-73.8648
C1_W3	New Haven CT	41.3083	-72.9279
C1_W4	Providence RI	41.8240	-71.4128
C1_W5	Boston MA	42.3601	-71.0589

C2_NJ_PHL (new)

waypoint_id	city	lat	lon
C2_W1	Newark NJ	40.7357	-74.1724
C2_W2	New Brunswick NJ	40.4862	-74.4518
C2_W3	Trenton NJ	40.2204	-74.7643
C2_W4	Philadelphia PA	39.9526	-75.1652

4. Authoritative Data Sources

3.1 Weather

- Source: Open-Meteo Weather Forecast API
- Granularity: Daily aggregates
- Forecast window: Next 2 forecast days (`forecast_days = 2`) to cover “next-day” planning
- Variables used:
 - `precipitation_sum` (mm/day)
 - `wind_gusts_10m_max` (km/h)
 - `temperature_2m_min` (°C)
- Hourly variables retrieved (optional reporting only; not used in risk score):
 - `temperature_2m`
 - `precipitation`
 - `wind_speed_10m`
 - `wind_gusts_10m`

3.2 Item Master Data

Source of truth for:

- `item_id`
- `item_name`
- medicine type
- packaging volume
- temperature requirements

5. Planning Horizon (48 hours) and Aggregation

The planning window covers **Day0** and **Day1** (48 hours). Weather is retrieved with `forecast_days = 2` .

5.1 Corridor risk aggregation across days (default)

Compute a corridor/day risk score first (0–3), then aggregate to a 48-hour corridor risk.

- **Day risk:** apply the waypoint trigger logic to each waypoint for that day; corridor day risk is the **max** waypoint score.
- **48-hour corridor risk:** take the **max** of Day0 and Day1 corridor day risk.

Teams may propose an alternative aggregation (e.g., weighted max) but must justify it in their report.

6. Weather Risk Rules (Operational Logic)

5.1 Weather Triggers (Daily Index)

Weather risk is triggered if any waypoint meets one or more conditions across the next 2 forecast days:

Condition	Open-Meteo Daily Variable	Threshold
Heavy Precipitation Risk	<code>precipitation_sum</code>	≥ 15.0 mm/day
High Wind Risk	<code>wind_gusts_10m_max</code>	≥ 45.0 km/h
Freezing Risk	<code>temperature_2m_min</code>	≤ 0.0 °C

5.2 Travel Time Buffer Policy (Score-based)

The `PlannerAgent` converts `risk_score_0_3` into a travel-time buffer:

<code>risk_score_0_3</code>	Travel Time Adjustment
0	No buffer
1	+10% buffer
2	+25% buffer
3	+40% buffer + escalation

7. Dispatch SLA Classes

SLA Tier	Medicine Category	Max Time-in-Transit
Tier 1	Life-critical	6 hours
Tier 2	Standard specialty	12 hours

Dispatch plans violating SLA must be flagged.

8. Truck Capacity & Packing Model

7.1 Truck Constraints

- Standard truck capacity: **10 volume units**
- Each `unique_item_id` is **1 volume unit**
- Packing inefficiency buffer: **+10%**
- Cold-chain items require **temperature-controlled trucks**

7.2 Volume Calculation

- $(\text{total_volume}) = \sum(\text{volume_per_unique_item_id})$
- $(\text{required_trucks}) = \lceil (\text{total_volume}) \times 1.10 / \text{truck_capacity} \rceil$

9. Medicine Master Reference (Item Truth Table)

This table is the authoritative mapping used for validation.

item_id	item_name	medicine_type	temp_control
10021	Remdesivir 100mg	Antiviral	Cold (2-8C)
10021	Remdesivir 200mg	Antiviral	Cold (2-8C)
10022	Insulin Lispro	Hormone	Cold (2-8C)
10035	Pembrolizumab	Monoclonal Antibody	Cold (2-8C)
10040	Epinephrine Auto-Injector	Emergency Drug	Room Temp (20-25C)
10050	Heparin Sodium	Anticoagulant	Room Temp (20-25C)
10060	Morphine Sulfate	Opioid Analgesic	Controlled Storage
10070	Albuterol Inhaler	Bronchodilator	Room Temp (20-25C)
99999	Experimental Oncology Drug	Clinical Trial Drug	Strict Cold Chain (-20C)

10. Shipment CSV Schema

Incoming shipment files must contain:

Column Name	Description
<code>shipment_date</code>	Dispatch day (used for Day0/Day1 comparisons)
<code>item_id</code>	Internal medicine identifier
<code>item_name</code>	Human-readable name
<code>unique_item_id</code>	Unit-level identifier
<code>dispatch_location</code>	Dispatch Hospital

Column Name	Description
corridor_id	Corridor key (required for multi-corridor planning)

Each row represents one physical unit.

11. Data Quality Rules (Anomaly Definitions)

Rule ID	Description	Action
DQ-01	Missing <code>unique_item_id</code>	Remove from the dispatch calculation
DQ-02	<code>item_id</code> not in master table	Flag for investigation
DQ-03	<code>item_name</code> mismatch for valid <code>item_id</code>	Flag for investigation
DQ-04	Duplicate <code>unique_item_id</code>	Flag for investigation

12. Exception Handling Policy

- Invalid rows are excluded from dispatch planning.
- All excluded rows must be:
 - Counted
 - Logged
 - Reported with reason codes

13. Resource Constraints and Allocation Policy (New)

Planning must account for limited resources shared across corridors and days.

13.1 Resource pools

Treat the following as scarce pools by day:

- `driver`
- `truck_standard`
- `truck_temp_controlled`

If a separate resource file is provided, that file is authoritative for daily availability.

13.2 Allocation objective (measurable)

When resources are insufficient, allocate to **minimize total penalty score** across corridors and days.

Use the following penalty model (per affected shipment unit):

Violation Type	Penalty per Unit
Tier 1 SLA violation	100 points
Tier 2 SLA violation	40 points
Cold-chain (temp-control) violation (any tier)	+80 points (in addition to SLA penalty, if any)
Non-SLA delivery delay (still within SLA, but not dispatched on requested day)	10 points

Total penalty score is:

- Sum all penalties for all impacted units across both days and all corridors.

The **allocation objective** is therefore:

- Choose a dispatch plan that yields the **lowest total penalty score**.
- If two plans tie on total penalty, prefer the plan with **fewer Tier 1 units impacted**.

This quantitative model makes corridor-level trade-offs explicit and gradable. Teams may propose adjustments, but any alternative must:

- Be defined with explicit numeric penalties/weights.
- Be applied consistently across corridors and days.
- Be documented in the final report (including rationale and example calculations).

14. Reporting Requirements

Final dispatch report must include:

- Weather risk summary (by corridor, and by day within the 48h window)
- Applied travel buffers
- Valid vs excluded shipment counts
- KPI comparison across corridors (shipment volume, excluded rate, Tier 1/Tier 2 mix)
- Recommended resource allocation (drivers/trucks/temp-controlled) by corridor and day, with rationale
- SLA risk flags

Appendix A — Item Master Appendix (Searchable)

This appendix is the operational reference for resolving missing/invalid identifiers and for standardizing analytics keys across item master variants.

A.1 Canonical Item Master (Authoritative)

Canonical keys are used for analytics and reconciliation. The shipment field `item_id` may not be globally unique (e.g., strength variants); use `canonical_item_id` as the stable key.

canonical_item_id	item_id	canonical_item_name	medicine_type	temp_control	product_class
RMD-100	10021	Remdesivir 100mg	Antiviral	Cold (2-8C)	Antiviral
RMD-200	10021	Remdesivir 200mg	Antiviral	Cold (2-8C)	Antiviral
INS-LIS	10022	Insulin Lispro	Hormone	Cold (2-8C)	Endocrine
PMB-KEY	10035	Pembrolizumab	Monoclonal Antibody	Cold (2-8C)	Oncology Biologic
EPI-AI	10040	Epinephrine Auto-Injector	Emergency Drug	Room Temp (20-25C)	Emergency
HEP-SOD	10050	Heparin Sodium	Anticoagulant	Room Temp (20-25C)	Anticoagulant
MOR-SUL	10060	Morphine Sulfate	Opioid Analgesic	Controlled Storage	Controlled
ALB-INH	10070	Albuterol Inhaler	Bronchodilator	Room Temp (20-25C)	Respiratory
EXP-ONC-CT	99999	Experimental Oncology Drug (Clinical Trial)	Clinical Trial Drug	Strict Cold Chain (-20C)	Clinical Trial
LEV-INH	10071	Levalbuterol Inhaler	Bronchodilator	Room Temp (20-25C)	Respiratory
INS-ASP	10023	Insulin Aspart	Hormone	Cold (2-8C)	Endocrine

A.2 Name Alias / Variant Table (Accepted)

Use this table when `item_name` contains typos/variants. Alias matches create a mapping suggestion with a confidence tier.

alias_name	canonical_item_id	confidence_tier	notes
Remdesivir 100 mg	RMD-100	ALIAS_MATCH	Space variant
Remdesivir 200 mg	RMD-200	ALIAS_MATCH	Space variant
Pembrolizumab (Keytruda)	PMB-KEY	ALIAS_MATCH	Brand in parentheses
EpiPen Auto Injector	EPI-AI	ALIAS_MATCH	Common brand phrasing
Heparin Na	HEP-SOD	ALIAS_MATCH	Abbreviation
Morphine Sulphate	MOR-SUL	ALIAS_MATCH	US/UK spelling

Albuterol Inhaler 90mcg	ALB-INH	ALIAS_MATCH	Dose suffix ignored for inhaler class
alias_name	canonical_item_id	confidence_tier	notes

A.3 Legacy / Deprecated / Invalid Identifier Mapping

Use this table when `item_id` is unknown/invalid in the master but can be reconciled to a canonical item. All legacy-mapped rows must be reported as “fixed via legacy mapping.”

legacy_item_id	canonical_item_id	rule	rationale
10020	RMD-100	LEGACY_ID_MAP	Vendor legacy ID for Remdesivir 100mg
20021	RMD-200	LEGACY_ID_MAP	Old system used 200xx for strength variants
1070	ALB-INH	LEGACY_ID_MAP	Truncated ID found in older CSV exports
99999	EXP-ONC-CT	SPECIAL_CASE	Clinical trial placeholder ID; requires strict cold chain

A.4 Substitution Policy (Operationally Allowed Cases)

Substitution is only allowed when explicitly permitted below. If not listed, do not substitute—flag for investigation.

canonical_item_id	substitute_canonical_item_id	allowed_when	not_allowed_when
ALB-INH	LEV-INH	Same medicine_type=Bronchodilator AND temp_control matches AND hospital approves interchange	Patient-specific order forbids substitution or formulary restriction applies
INS-LIS	INS-ASP	Cold-chain endocrine hormones only; substitution requires pharmacist approval	Patient-specific shipment or dosing regimen differs

A.5 Identifier Format Rules (`unique_item_id`)

These patterns help validate and (when safe) regenerate missing `unique_item_id` values. If regeneration is used, mark as `generated_identifier` and log the basis.

product_class	expected_regex	example	notes
Antiviral	<code>^RMD-\d{4}-\d{4}\$</code>	RMD-2026-0042	Year + 4-digit sequence
Oncology Biologic	<code>^PMB-\d{4}-\d{5}\$</code>	PMB-2026-00017	Year + 5-digit sequence
Emergency	<code>^EPI-\d{4}-\d{4}\$</code>	EPI-2026-0101	Year + 4-digit sequence
Controlled	<code>^CTRL-\d{4}-\d{6}\$</code>	CTRL-2026-000123	Year + 6-digit sequence
Respiratory	<code>^INH-\d{4}-\d{5}\$</code>	INH-2026-00234	Year + 5-digit sequence
Clinical Trial	<code>^CT-\d{4}-[A-Z0-9]{6}\$</code>	CT-2026-A1B2C3	Randomized suffix

A.6 Decision Rules (How to Use This Appendix)

- **D1 Precedence:** If `unique_item_id` is missing, follow DQ-01 unless an explicit regeneration rule (A.5) is permitted and logged.
- **D2 Canonical key:** Reconcile to `canonical_item_id` for analytics; preserve raw fields in logs when possible.
- **D3 Exact match:** If (`item_id`, `item_name`) maps to a single canonical row in A.1, confidence = `EXACT_MATCH`.
- **D4 Alias match:** If `item_name` matches A.2, map to `canonical_item_id`, confidence = `ALIAS_MATCH`.
- **D5 Legacy match:** If `item_id` matches A.3, map to `canonical_item_id`, confidence = `LEGACY_ID_MAP`.
- **D6 Conflicts:** If `item_id` points to multiple canonical items, use `item_name` to disambiguate; otherwise flag `UNRESOLVED_CONFLICT`.
- **D7 Substitution:** Only substitute when listed in A.4 and approval condition is met; otherwise flag (no auto-fix).
- **D8 Reporting:** Every modified row must include a reason code: `exact_match`, `alias_match`, `legacy_id_map`, `substituted`, `generated_identifier`, `excluded_unresolved`.